

Reading the City's Fever: A Cross-Disciplinary Sensor Mesh for Urban Heat Island Detection

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Abstract. Urban heat islands (UHIs) are typically studied either through sparse fixed-station meteorology or through remote-sensing imagery that misses fine-grained, street-level variation. This paper reports on a nine-month deployment of a low-cost, solar-powered sensor mesh across four neighborhoods of the fictional city of Meridian Bay, combined with a lightweight edge-to-cloud pipeline and a policy-facing dashboard co-designed with municipal planners. We show that intra-neighborhood temperature differentials reach 3.8°C on summer afternoons, that tree-canopy fraction is the strongest single predictor of cooling, and that translating raw sensor output into planning-relevant tiers required iterative input from environmental engineers, computer scientists, and policy analysts working as a single team rather than in sequence. We argue that this three-way loop, not the sensor hardware itself, is the paper's main contribution, and we offer a template for symposium tracks that bring these communities together.

Keywords: urban heat islands, low-cost sensing, edge computing, interdisciplinary methods, environmental policy

Key Contributions

- A 42-node solar sensor mesh built from off-the-shelf components, validated against two reference-grade weather stations ($R^2 = 0.94$).
- An edge-aggregation pipeline that reduces uplink volume by 87% without discarding peak-hour resolution.
- A four-tier confidence scoring scheme that lets non-technical planners read sensor output without a statistics background.
- A reflection on where three-discipline collaboration broke down and what fixed it, offered as a working note for other symposium tracks.

1 Introduction

Cities warm unevenly. A parking lot and a shaded park three blocks apart can differ by several degrees at the same hour, yet the instruments that most municipal climate offices rely on – one or two airport-grade weather stations – cannot see that difference at all. Remote sensing fills part of the gap, but satellite thermal bands report surface temperature, not the air temperature that determines whether a resident on a heat-advisory day is actually at risk.

The Meridian Bay Urban Resilience Office asked our three groups – an environmental engineering lab, a pub-

lic policy program, and a computer science department – to jointly answer a deceptively simple question: *which blocks need cooling infrastructure first, and how confident should the answer be?* Answering it required a sensor network cheap enough to deploy at block-level density, a computing pipeline that could turn a firehose of noisy readings into something stable, and a policy framing that a city council member could act on inside a single meeting. No one discipline could deliver all three.

This short paper documents the resulting system and,